DIMENSION

The Basis Size Theorem

Every basis of V has the same number of elements.

Two bases

$\mathcal{B}$ with n elements, $\mathcal{C}$ with m

- In a space spanned by n vectors, more than n vectors are dependent
- A spanning set has at least as many vectors as an independent set

$\mathcal{B}$ SPANS, $\mathcal{C}$ IS INDEPENDENT →

First inequality

$m \leq n$

SWAP THE ROLES →

Second inequality

$n \leq m$, so $m = n$

- Dimension is well-defined